



Department of Mechanical, Industrial, and Mechatronics Engineering

Milestone 3 Report

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Milestone 3

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Summary:

This project involved the design and testing of an analog ruler on a slider with a rotary display. The system used an optical distance sensor to detect target position, a DC gearmotor to move the pointer, a motor driver to power the motor, as well as the NI USB-6001 for data acquisition and control. So that the mechanical and electrical parts could actually work together a mechanical base, pointer, dial, and potentiometer linkage were built. A closed loop control was used so the pointer could respond to changes in position as well as reject disturbances with reasonable stability, and calibration was completed to relate measured distance to pointer angle. The final device worked well overall, even though noise, alignment, and integration between the parts like the sensor, motor, potentiometer, and control program made the project harder. We learned that combining several subsystems into one working design with feedback is challenging.

Introduction:

This project focused on the design and development of an analog ruler on a slider with a rotary display. The main idea was to measure the position of a target and then show that position using a rotating pointer on a dial, so that the final system could act like a mechanical style measuring device with electronic control. The project was not only about making the pointer move but also about building a full system that had a proper mechanical structure and electrical hardware, calibration procedure, with feedback control working together.

The system used an optical distance sensor to measure the distance to a target, and this measurement was then used as the main input to the controller. A DC gearmotor was used to rotate the pointer, while a DRV8833 motor driver was used to supply and control the motor. For data acquisition and control signal generation, the NI USB-6001 DAQ was used. It provided the interface between the sensor signals and control program, as well as the motor actuation. A potentiometer was also included in the system so the rotary position could be monitored as part of the feedback loop. In simple terms, the sensor measured the target position, the controller decided how the motor should respond, and the motor then moved the pointer to the required angle. PWM control was used to vary the effective input to the motor, which made controlled motion possible instead of just turning the motor fully on or fully off.

The mechanical side of the project was just as important as the electrical side. A base had to be designed so the motor and other parts could be mounted properly, and the dial, pointer, with the 3D printed shaft connection to the potentiometer had to be made in a way that was stable and usable. This part seems simple at first, but it actually mattered a lot because even small alignment problems could affect the measurement and the pointer response. Based on the earlier project milestones, the design also had to show a complete mechanical system, proper

calibration work, and a working feedback control setup with a clear flowchart, wiring diagram, and LabVIEW VI.

The main objective of this project was to design, build, and calibrate, as well as test a working analog ruler with rotary readout that could respond to measured distance and reject disturbances with reasonable stability. Another major goal was to integrate the different subsystems into one final design, including the sensor, motor, motor driver, potentiometer, DAQ, and control program. This was honestly one of the more difficult parts of the project, because each part could work on its own, but getting them all to work together properly was harder than expected. Overall, the project was meant to apply concepts from measurements, instrumentation, sensing, calibration, and feedback control in one complete device rather than in separate small exercises.

Mechanical Design:

Design Conditions

For the system it was crucial that the mechanical design effectively supported all components with a stable structure, housing all the components with easy to assemble parts. The design layout was aimed to fully enclose the DAQ, breadboard, sensor, motor, and potentiometer within the base. The dial was featured on the side of the base to easily see the measurement results. Furthermore, couplers were designed in order to transmit the torque from the motor to both the potentiometer as well as the dial pointer.

The dial pointer was designed to be narrow from the tip in order to point to the markings on the dial without blocking numbers or being difficult to read.

The main housing consists of a rectangular base with openings on opposite sides for easy access to the inside as well as housing for the sensor. Furthermore, one side of the enclosure features a slit for a slider mechanism, this is used in order to assist the sensor to measure the distance between the object and the sensor. It provides an easy and clear way to show where the object is located by moving the slider to the desired position of the object location.

Internal layout & Design Changes

Inside the housing the motor is mounted on a rectangular platform raised above the surface in order to properly allow the rotary dial to be mounted properly. In design iteration 1, a design issue that arose was the rotary dial display hanging below the housing; this required the need to hang the dial off the edge in order for the structure to be stable. In the next iteration, the motor was raised in order to fix this problem. Moreover, another design change from iteration 1 was the introduction of a potentiometer mount in order to fully support the potentiometer and prevent movement or shifting during motor operation, which was a critical change from the first prototype to ensure consistent feedback signals.

Torque Transmission and Couplers

The motor is attached to both the potentiometer and the rotary display using two different couplers. Both couplers were designed to provide a tight fit and caused no slippage from occurring, this allowed for smooth and consistent motion.

The dial coupler consisted of a joint assembly of the pointer used on the display as well as the coupler for the motor connection; this was done to provide a stable connection from the motor to the display pointer and allowed for no slippage to occur during motor operation. Furthermore the coupler between the motor and potentiometer transmit the torque without slippage, ensuring the angular position measured by the DAQ accurately reflects the pointer's location.

Detailed CAD drawings as well as initial sketches are found in the Appendix.

Electrical System and Data Acquisition:

Electrical Design:

An electrical system is responsible for generating, distributing, and utilizing electrical power to all the components, to perform the desired work.

The electrical system for this project was responsible for:

- Powering the optical sensor and motor
- Measuring the optical sensor and potentiometer signals
- Sending control signals from the DAQ
- Interfacing the controller with the motor driver

Components	Function
SHARP Optical Sensor [2]	Uses infrared light to detect distance of object
DC Gearmotor [3]	Rotates pointer to desired angle
Motor Controller [4]	Allows the DAQ to drive the motor safely
NI USB-6001 DAQ [5]	Reads analog signals and sends control signals
Potentiometer	Measures pointer angle
Breadboard + Jumper Wires	Physically connects components together
9v Power Supply	Provides electrical power
Breadboard Power Distribution Board	Steps voltage down to 5v

Table 1. List of Components

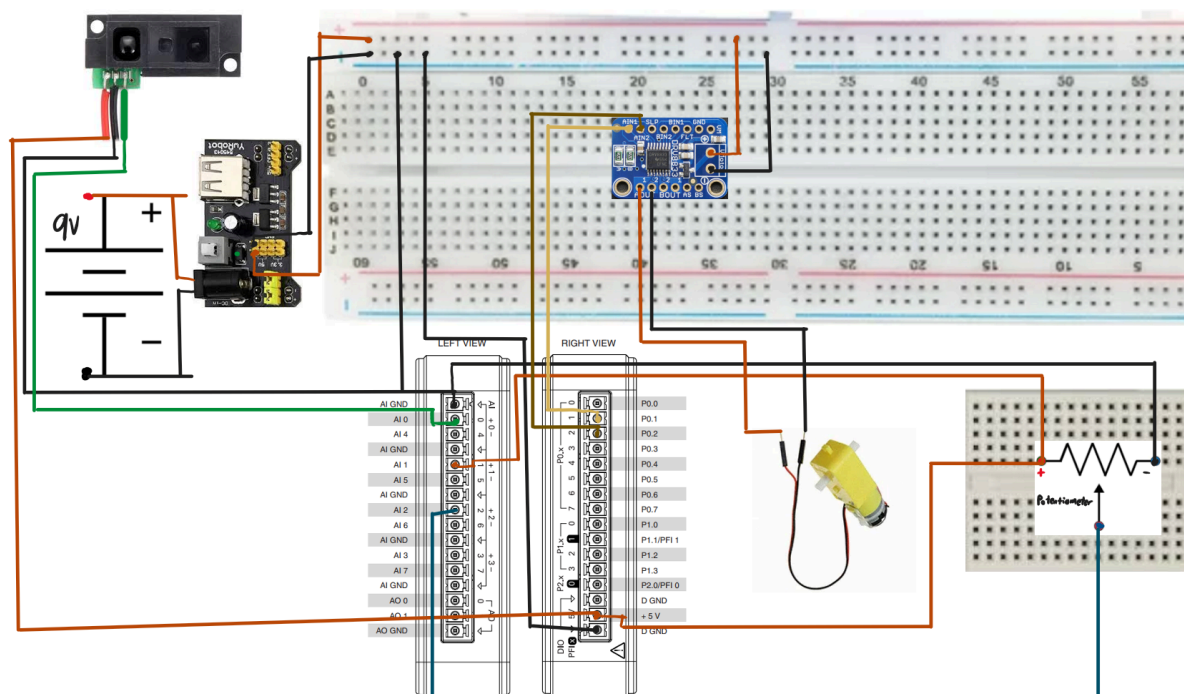


Figure 1. Wiring Diagram

The optical sensor was powered through the DAQ, hence the wiring to +5V and AI GND. Its analog output voltage was connected to the analog input channel AI0 on the DAQ so that the distance measured by the sensor can be used in Labview.

The potentiometer was mechanically linked to the DC gearmotor, which in turn is linked to the pointer. Similar to the optical sensor, the DAQ supplies a reference voltage across the

potentiometer, hence it has the same wiring to +5V and AI GND. The output voltage was connected to a different analog input channel AI2 on the DAQ and used as the feedback signal for pointer position.

The DC gearmotor on the other hand was powered by the power supply and driven by the motor controller, hence it was connected to AOUT1 and AOUT2 on the motor controller. The power supply was connected to VM and GND of the motor controller to supply electrical power to the DC gearmotor. The motor controller acted as an interface between the low-power DAQ output signals and the motor power circuit. This allowed direction and actuation commands to be sent safely.

All components shared a common ground, hence the additional wirings from AI GND and D GND of the DAQ to the common ground.

Control System:

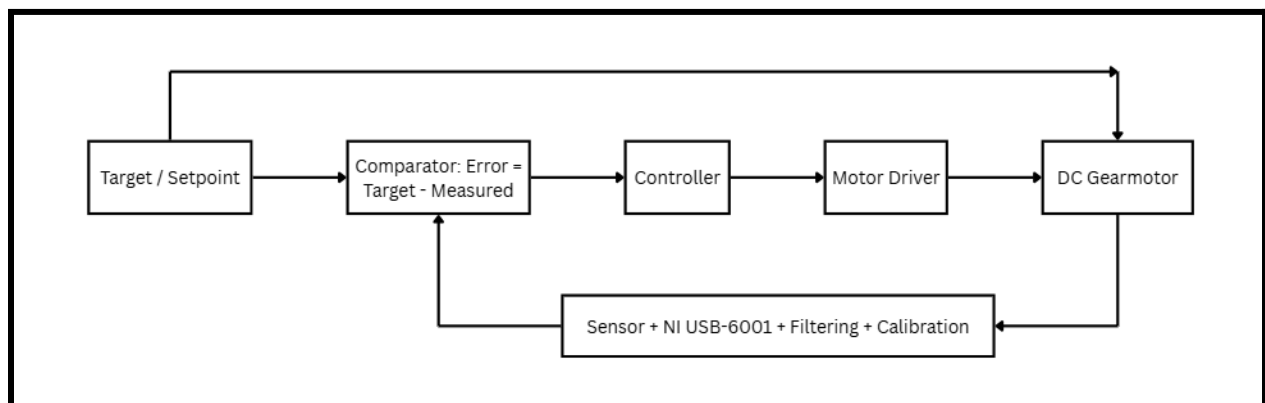


Figure 2: Block diagram of the control system of this LabVIEW program.

The control system was designed as a closed-loop feedback system in which the desired pointer position acts as the setpoint and the measured system response is fed back for comparison. The sensor signal is acquired through the NI USB-6001, then filtered and calibrated so that the raw voltage can be converted into a usable measured value. This measured signal is compared with the target value to generate an error, which is then processed by the controller to produce a control command. The command is sent through the motor driver to the DC gearmotor, which rotates the pointer to the required position. By continuously feeding the measured response back into the system, the controller is able to reduce error and correct for disturbances, making the overall design more accurate and stable than an open-loop system.

LabVIEW Program Description:

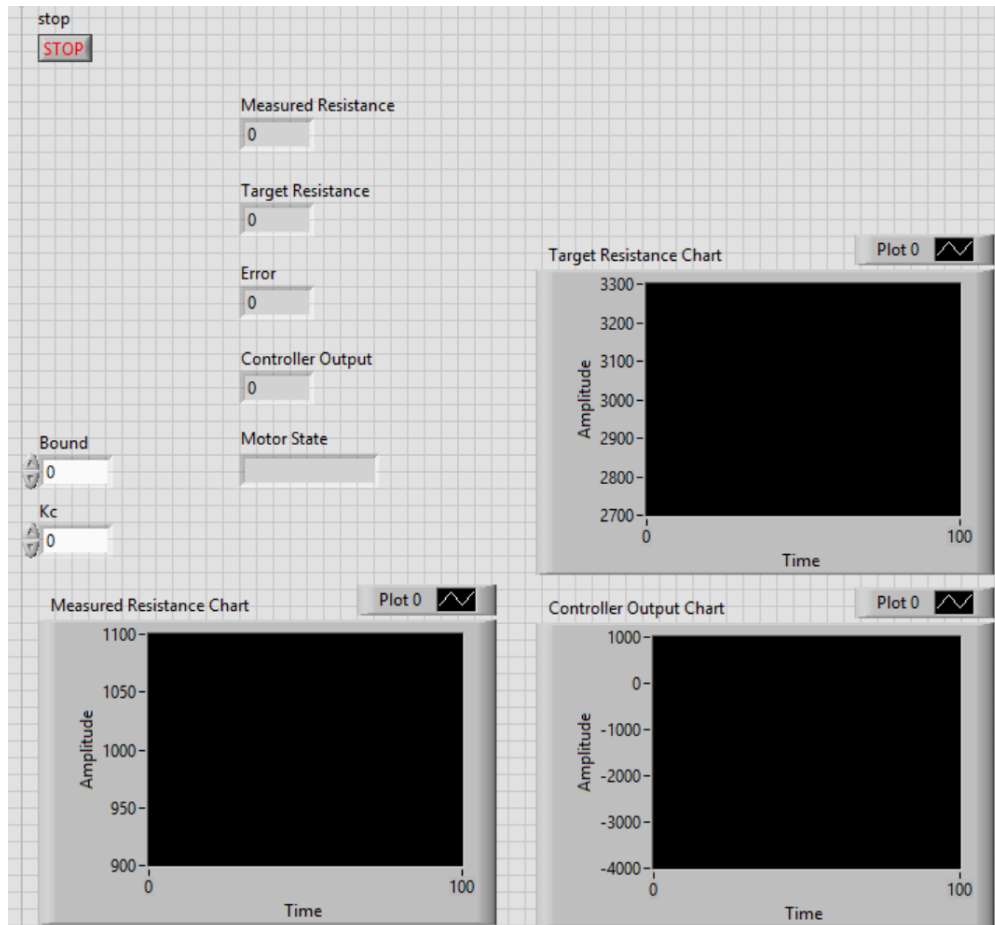


Figure 3. Main Control VI Front Panel

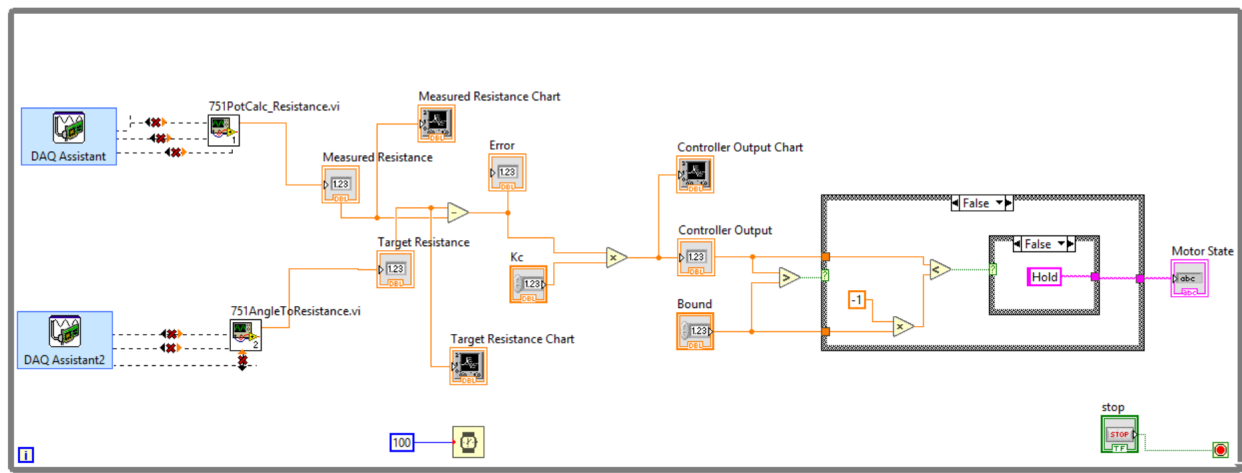


Figure 4. Main Control VI Block Diagram

- Note: Due to LabVIEW Installation errors, the DAQ Assistants are not properly connected.

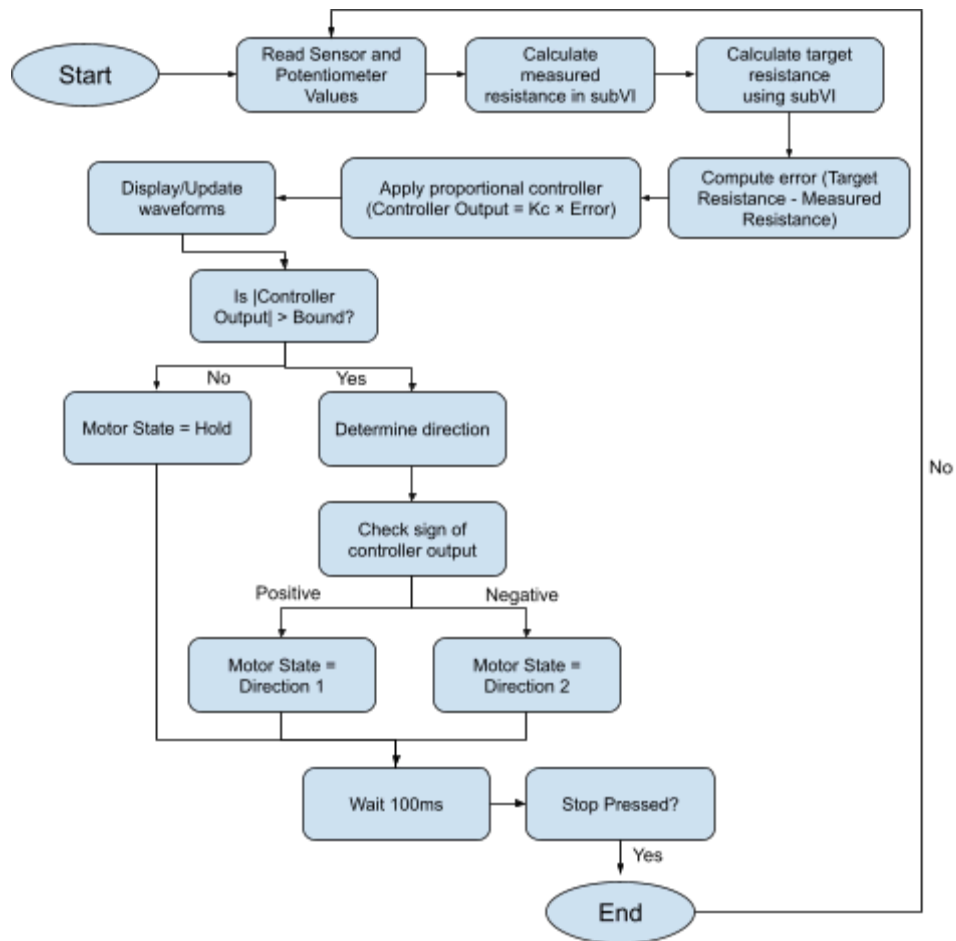


Figure 5. Main VI Flowchart

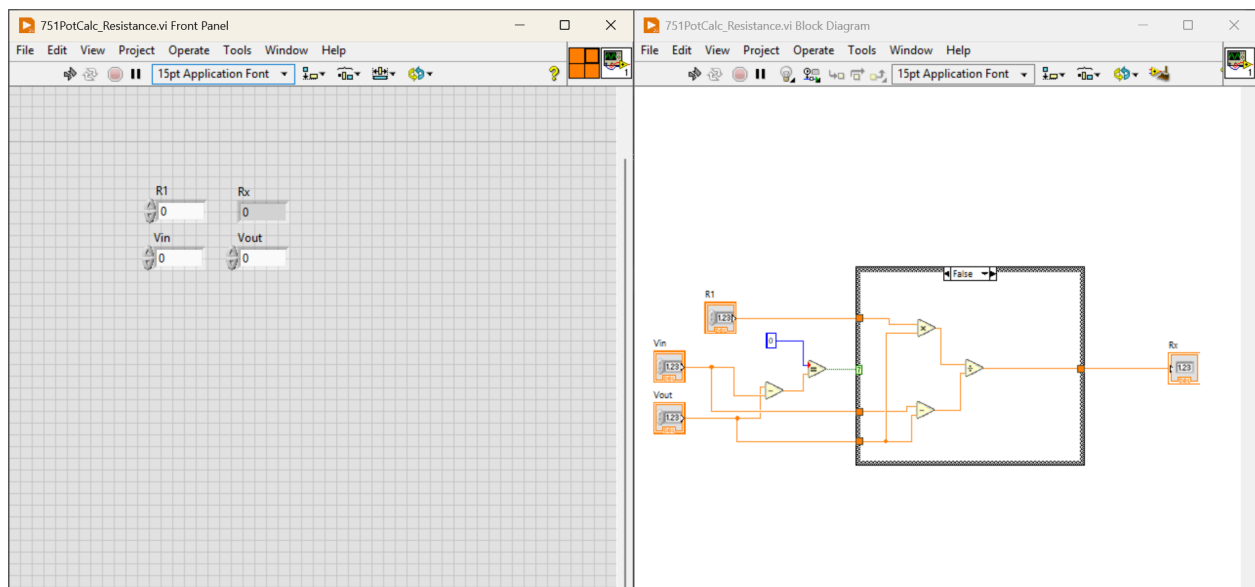


Figure 6. PotCalc_Resistance Sub VI

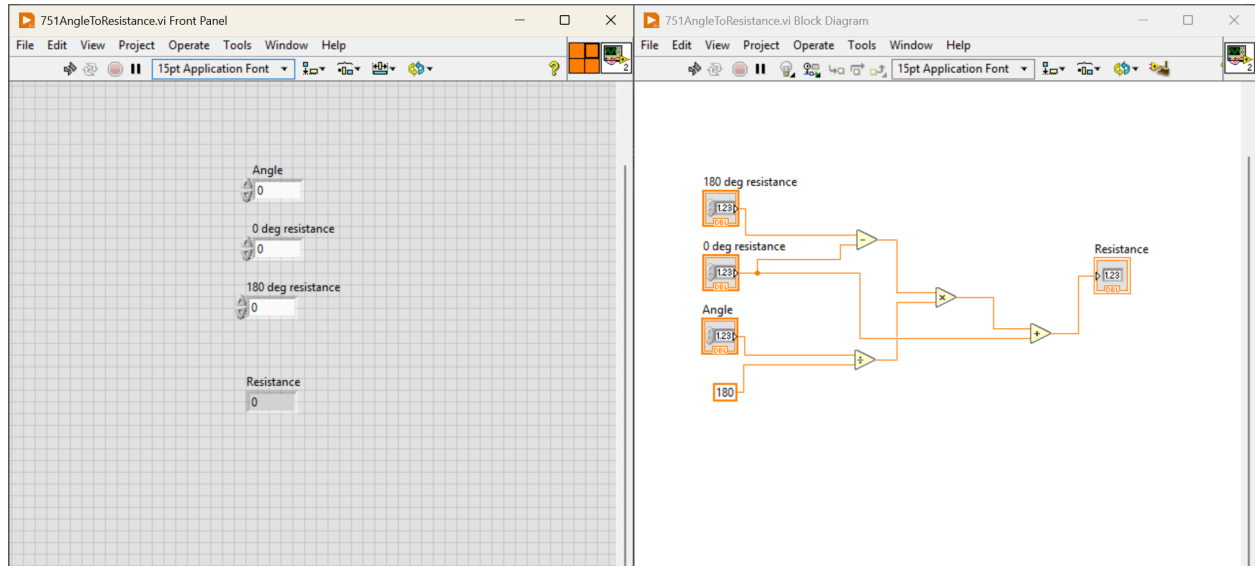


Figure 7. AngleToResistance Sub VI

1. Sensor Data Acquisition and Calibration

This section of the LabVIEW program is responsible for acquiring the sensor signal through the NI USB-6001 and converting it into a usable feedback variable for the control system. The analog voltage produced by the sensor is read continuously inside the main loop and then processed using the calibration relationship obtained experimentally. This allows the raw sensor output to be converted into a meaningful physical quantity, such as target position or equivalent pointer reference. As a result, the controller operates on calibrated measurement data rather than raw voltage values, which improves the accuracy and interpretability of the system response.

2. Error Calculation and Feedback Control

Once the measured signal has been obtained, it is compared with the desired pointer response to calculate the control error. This error is defined as the difference between the target value and the measured value and represents how far the system is from the required response at any given instant. The error signal is then processed by the controller to generate a control effort that drives the motor in the direction needed to reduce the mismatch. In the present implementation, the control law is proportional, meaning that the controller output is directly related to the magnitude of the error. This forms the core of the feedback loop and allows the system to continuously correct itself during operation.

3. Motor Command Generation and Actuation

The controller output is next converted into a motor command that determines how the rotary pointer should move. The sign and magnitude of the control signal are used to decide whether the motor should rotate forward, reverse, or remain in a hold state. A threshold value is

used so that small control efforts near the setpoint do not cause unnecessary switching or oscillation. Once the motor state has been determined, the corresponding command is sent through the NI USB-6001 to the motor driver, which supplies the required actuation to the DC gearmotor. The motor then rotates the pointer until the error is reduced.

4. Program Monitoring and Execution

The entire LabVIEW program operates inside a continuous while loop so that data acquisition, control calculations, and motor commands are updated repeatedly in real time. A timing block is included to regulate the loop speed and ensure stable execution, while a stop control allows the program to be ended safely by the user. During execution, the front panel displays key variables such as the measured response, target response, error, controller output, and motor state. Charts are also used to visualize system behaviour over time, which is useful for controller evaluation, debugging, and discussion of performance in the report.

Results and Evaluation:

Sensor Calibration:

The following procedure was followed for sensor calibration:

1. The target object was placed at known distances from the sensor.
2. The object was moved across the required operating range in fixed increments.
3. At each position, the sensor output voltage was measured using the VI in Labview.
4. Multiple readings were taken at each distance to reduce random error.
5. The measured voltage was recorded along with the true distance.
6. A calibration plot of voltage versus distance, or distance versus voltage, was generated.
7. A best-fit equation was obtained and used to convert sensor voltage into distance during system operation.
8. The calibration was checked for nonlinearity, repeatability, and hysteresis if tested.

Sensor	Cycle 1 (V)	Cycle 2 (V)	Cycle 3 (V)	Sensor Cycles Mean Voltage
2cm	1.83871	1.84257	1.80399	1.828423333
3cm	1.431	1.46187	1.45029	1.44772
4cm	1.18406	1.16734	1.14548	1.165626667
5cm	0.980847	0.976989	0.974416	0.9774173333
6cm	0.817505	0.838084	0.836797	0.8307953333
7cm	0.722329	0.723616	0.721043	0.7223293333
8cm	0.632298	0.62844	0.643874	0.6348706667
9cm	0.552557	0.57828	0.569277	0.5667046667
10cm	0.488249	0.486963	0.490821	0.4886776667
11cm	0.43423	0.43423	0.457381	0.441947
12cm	0.377639	0.393073	0.398218	0.3896433333
13cm	0.33391	0.353202	0.354488	0.3472
14cm	0.291467	0.306901	0.306901	0.3017563333
15cm	0.278605	0.273461	0.272174	0.2747466667
14cm	0.294039	0.313331	0.313331	0.3069003333
13cm	0.336482	0.353202	0.354488	0.3480573333
12cm	0.373781	0.389215	0.396932	0.3866426667
11cm	0.43423	0.432944	0.436803	0.434659
10cm	0.489535	0.47796	0.493393	0.4869626667
9cm	0.55127	0.530692	0.549984	0.543982
8cm	0.62844	0.631012	0.627154	0.6288686667
7cm	0.721043	0.741622	0.722329	0.7283313333
6cm	0.818791	0.836797	0.838084	0.831224
5cm	0.97313	0.97313	0.971844	0.9727013333
4cm	1.16091	1.14676	1.16605	1.157906667
3cm	1.45029	1.44772	1.44644	1.44815
2cm	1.82199	1.82328	1.82199	1.82242

Table 2: Measured Distance and Voltage Relationship of Sensor

The FSO can be determined as the highest mean voltage,

$$FSO = V_{mean_{MAX}} = 1.828423333 V$$

Sensor Cycles Mean vs. Sensor Distance

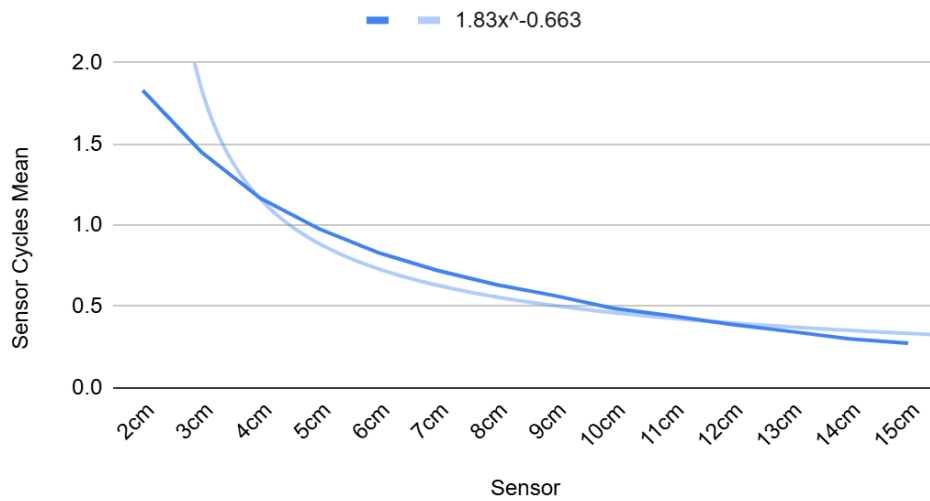


Figure 8. Sensors Cycles Mean vs Sensor Distance Plot

Hysteresis Error

Sensor	Hysteresis Error
2cm	0.006003333
3cm	0.00043
4cm	0.00772
5cm	0.004716
6cm	0.0004286667
7cm	0.006002
8cm	0.006002
9cm	0.0227226667
10cm	0.001715
11cm	0.007288
12cm	0.0030006666
13cm	0.0008573333
14cm	0.005144
15cm	0
14cm	0.005144
13cm	0.0008573333
12cm	0.0030006666
11cm	0.007288

10cm	0.001715
9cm	0.0227226667
8cm	0.006002
7cm	0.006002
6cm	0.0004286667
5cm	0.004716
4cm	0.00772
3cm	0.00043
2cm	0.006003333

Table 3: Hysteresis Error

$$E_h = \frac{\max|V_{up,mean} - V_{down,mean}|}{FSO} \times 100\%$$

The largest hysteresis difference is at 9 cm:

$$\Delta V_{h,max} = 0.0227226667 \text{ V}$$

$$E_h = \frac{0.0227226667}{1.828423333} \times 100 = 1.24\%$$

$$\text{Hysteresis Error} = 1.24\% \text{ FSO}$$

Repeatability Error

Sensor	Repeatability Error
2cm	0.03858
3cm	0.03087
4cm	0.03858
5cm	0.006431
6cm	0.020579
7cm	0.002573
8cm	0.015434
9cm	0.025723
10cm	0.003858
11cm	0.023151
12cm	0.020579
13cm	0.020578
14cm	0.015434
15cm	0.006431
14cm	0.019292
13cm	0.018006
12cm	0.023151
11cm	0.003859
10cm	0.015433
9cm	0.020578
8cm	0.003858
7cm	0.020579
6cm	0.019293
5cm	0.001286
4cm	0.01929
3cm	0.00385
2cm	0.00129

Table 4: Repeatability Error

$$E_r = \frac{\max(V_{max} - V_{min})}{FSO} \times 100\%$$

The largest spread is:

$$\Delta V_{r,max} = 0.03858 \text{ V}$$

$$E_r = \frac{0.03858}{1.828423333} \times 100 = 2.11\%$$

$$\text{Repeatability Error} = 2.11\% \text{ FSO}$$

Linearity Error:

Sensor	Linearity Error
2cm	1.825421667
3cm	1.447935
4cm	1.161766667
5cm	0.9750593333
6cm	0.8310096667
7cm	0.7253303333
8cm	0.6318696667
9cm	0.5553433334
10cm	0.4878201667
11cm	0.438303
12cm	0.388143
13cm	0.3476286667
14cm	0.3043283333
15cm	0.2747466667
14cm	0.3043283333
13cm	0.3476286667
12cm	0.388143
11cm	0.438303
10cm	0.4878201667
9cm	0.5553433334
8cm	0.6318696667
7cm	0.7253303333
6cm	0.8310096667
5cm	0.9750593333
4cm	1.161766667
3cm	1.447935

2cm	1.825421667
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Table 5: Linearity Error

$$E_L = \frac{\max |V_{measured} - V_{BFSL}|}{FSO} \times 100\%$$

Using the averaged calibration points, the best-fit straight line is:

$$V_{BFSL} = -0.10314 d + 1.61913$$

The deviation from that line at each distance:

$$|V_{avg} - V_{BFSL}|$$

So at 2 cm:

Predicted by line:

$$V_{BFSL}(2) = -0.10314(2) + 1.61913 = 1.41286 V$$

Measured averaged value:

$$V_{avg}(2) = 1.825421667 V$$

The largest deviation is:

$$|1.825421667 - 1.41286| = 0.41256 V$$

$$\Delta V_{L,max} = 0.41256 V$$

$$E_L = \frac{0.41256}{1.828423333} \times 100 = 22.56\%$$

$$Linearity Error = 22.56\% FSO$$

Overall Instrument Error

$$E_{overall} = E_L + E_h + E_r$$

Substitute values:

$$E_{overall} = 22.56 + 1.24 + 2.11$$

$$E_{overall} = 25.91\%$$

$$25.91\% FSO$$

In volts:

$$0.2591 \times 1.828423333 = 0.4739 \text{ V}$$

Final answer:

$$\pm 25.91\% \text{ FSO or } \pm 0.4739 \text{ V}$$

Dynamic Calibration errors for Potentiometer:

We know dynamic angle error is defined as

$$\theta_{error}(t) = \theta_{out}(t) - \theta_{in}(t)$$

$\theta_{in}(t)$ = commanded angle from the sensor

$\theta_{out}(t)$ = actual angle from the potentiometer

From Figure 8 we get

$$V_s(d) = 1.83 d^{-0.663}$$

V_s = sensor voltage

d = distance in cm

$$V_s = 1.83 d^{-0.663}$$

$$\frac{1.83}{V_s} = d^{0.663}$$

$$d(V_s) = \left(\frac{1.83}{V_s} \right)^{1.508}$$

Now we can convert distance to dial angle so we used 2 cm to 15 cm mapped to 0° to 130°. So

$$\frac{d-2}{15-2} = \frac{\theta-0}{130-0}$$

$$\theta(d) = 130 \left(\frac{d-2}{13} \right)$$

$$\theta(d) = 10(d - 2)$$

Now we can just substitute $d(V_s)$ into $\theta(d)$:

$$\theta_{in}(V_s) = 10 \left[\left(\frac{1.83}{V_s} \right)^{1.508} - 2 \right]$$

For the potentiometer, the relation is linear so we can find

$$\frac{V_p - V_{min}}{V_{max} - V_{min}} = \frac{\theta_{out} - 0}{130 - 0}$$

$$\theta_{out}(V_p) = 130 \left(\frac{V_p - V_{min}}{V_{max} - V_{min}} \right)$$

$$\frac{V_p - 1.07512}{4.48558 - 1.07512} = \frac{\theta_{out}}{130}$$

$$\frac{V_p - 1.07512}{3.41046} = \frac{\theta_{out}}{130}$$

$$\theta_{out} = 130 \cdot \frac{V_p - 1.07512}{3.41046}$$

$$\theta_{out}(V_p) = 38.11 (V_p - 1.07512)$$

Final dynamic error equation:

$$\theta_{error}(t) = 38.11 (V_p(t) - 1.07512) - 10 \left[\left(\frac{1.83}{V_s(t)} \right)^{1.508} - 2 \right]$$

Discussion

The accurate measurement of the potentiometer resistance is heavily influenced by the noise generated by the mechanical and electrical systems. More specifically, from design iteration 1, noise was caused by slippage between the motor and potentiometer coupler, this caused values to not be registered when calibrating the system. Moreover, the motor transmits vibration directly to the display, and since torque is being transmitted through couplers, it is another source of noise for the system. Another way to combat this problem was to add mounts for both the motor and the potentiometer, to ensure both electronic components do not move or have and slippage through the coupler, the display was also raised and mounted with a combined dial & coupler to ensure no slippage occurs there as well. Furthermore, noise from mechanical components can further be reduced by using dampening materials in order to provide more support.

Electrical noise can be reduced by simplifying wiring, and reducing lengths between wires and components. Moreover, the motor showed further inaccuracies when tracking values, using PWM as well as a closed loop control system, utilizing the proportional gain to reduce steady state error and aid in increasing the responsiveness of the system. The sensor also provided significant inaccuracies, this was resolved by implementing a low-pass filter in order to cut off frequencies higher than what was required and reduce the noise of the sensor, removing any fluctuations and spikes in values.

By comparing real-world results to the expected theoretical behaviour, the results show the difficulty caused by disturbances and errors from electrical components. Errors such as motor speed, calibration errors or issues caused by the LabVIEW system proved to cause discrepancies in results.

The pointer showed discrepancy between the actual value and the value read due to motor behaviour and physical limitations, this was due to stall voltage, motor friction. Moreover, the precision of the results was heavily influenced by the sensor as well, causing discrepancies with noise as well as the quality of the sensor. The sensor showed the tendency to create random spikes when used, as mentioned above, these spikes were reduced by the implementation of a low-pass filter.

Moreover, the potentiometer also caused errors in the results expected, the potentiometer features a non-linear region of values close to 0V, the readings of low distance values caused nonlinearity error due to the property of the potentiometer.

Overall, the final results showed errors when compared to theoretical expectations, despite these small errors, the system demonstrates high repeatability which validates the sensors capabilities as well as the overall system performance.

Conclusion:

Overall, this project was successful in showing us how an analog ruler on a slider with a rotary display can be designed. Everything from how it gets built, then calibrated, and even controlled as one full system. The final design used an optical distance sensor to measure the target position, a DC gearmotor and motor driver to move the pointer, and the NI USB-6001 for data acquisition and control. Even though the final device worked well, it was all because of how the feedback control, the mechanical structure, sensing, and calibration all worked together properly, rather than just making one part work. Taking the time to implement everything one by one, including even the smallest parts, such as the organized wiring, allowed for easier troubleshooting and was ultimately the key to finding success in this project.

As a result of doing this project we learned that one of the biggest things was that real engineering always comes with tradeoffs. A design can seem fine in theory at the beginning, but noise, alignment, wiring, and mechanical limitations can still affect the final result a lot more than expected. Another thing we should have accounted for is the number of broken/faulty parts we would discover in the process. If this project were to be done again, more time would be spent on the mechanical design at the beginning, and each subsystem would also be tested more carefully before everything was put together into one final setup because it would have likely reduced some of the problems we would face later on. The performance of the design could have been improved by reducing noise in the sensor readings, making the mechanical setup more precise, and refining the calibration process with more data points so the relationship

between measured distance and pointer angle could be more accurate. Overall, this project helped show us that combining different subsystems with feedback into one working design is probably one of the hardest parts, even if each part works fine by itself.

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Appendix

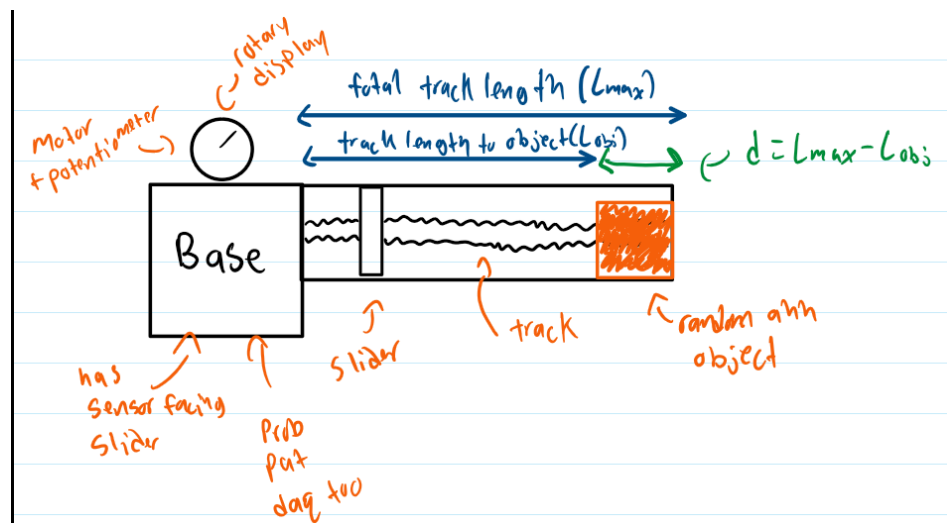


Figure 9 Initial Sketch of System

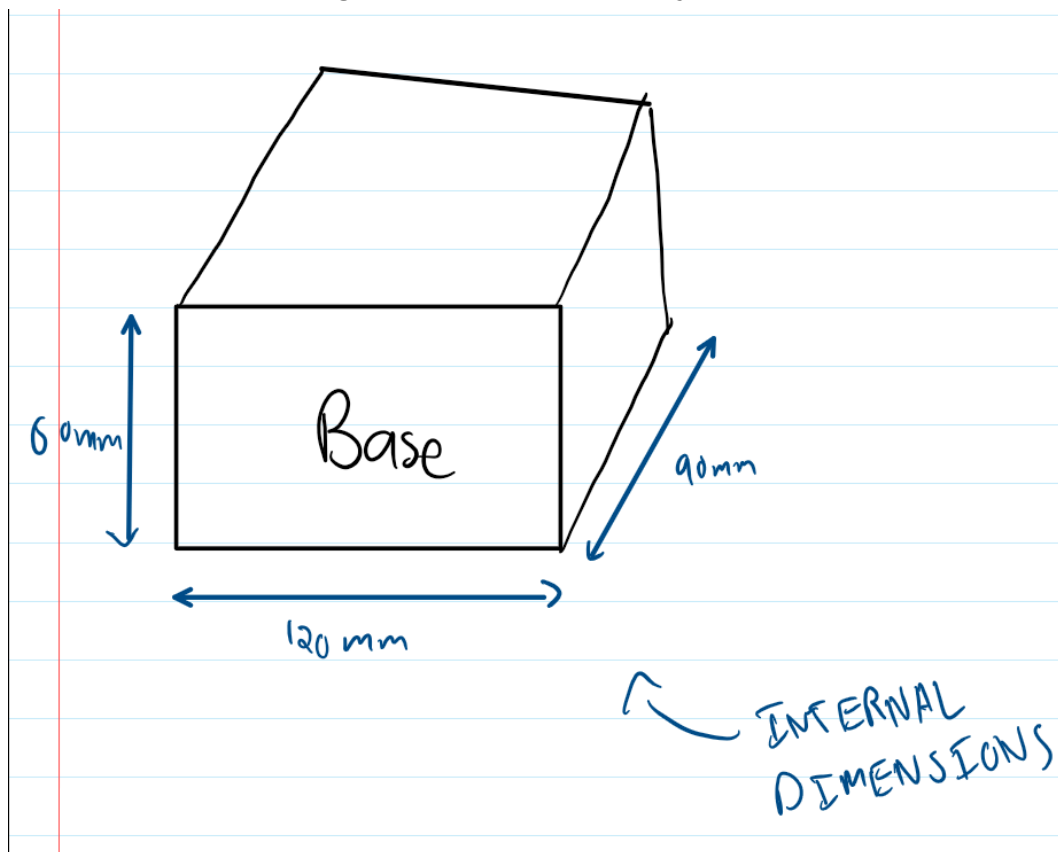


Figure 10 Initial Base Dimensions

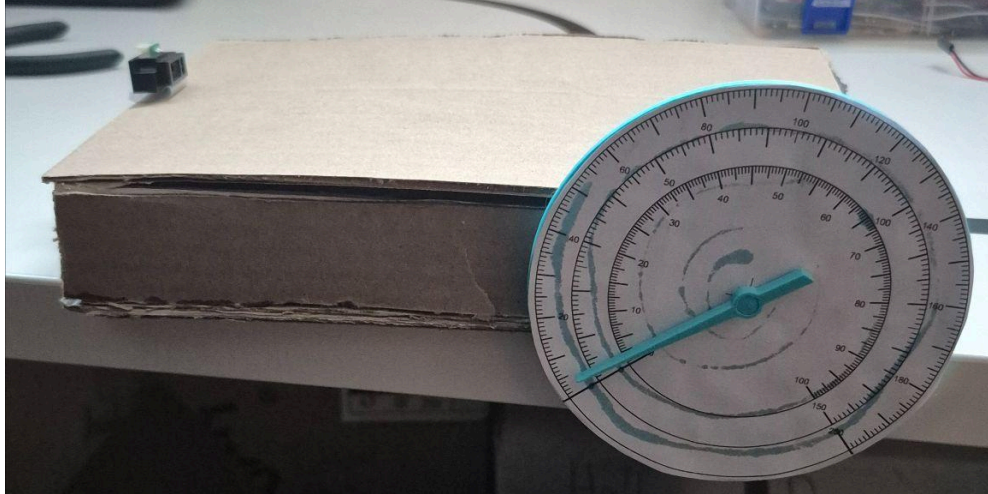


Figure 11 Design Iteration 1

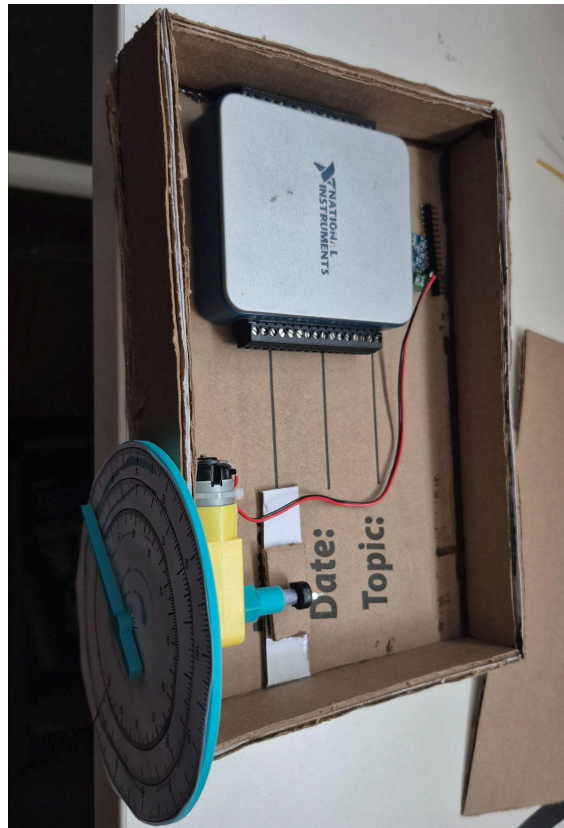


Figure 12 Design Iteration 1

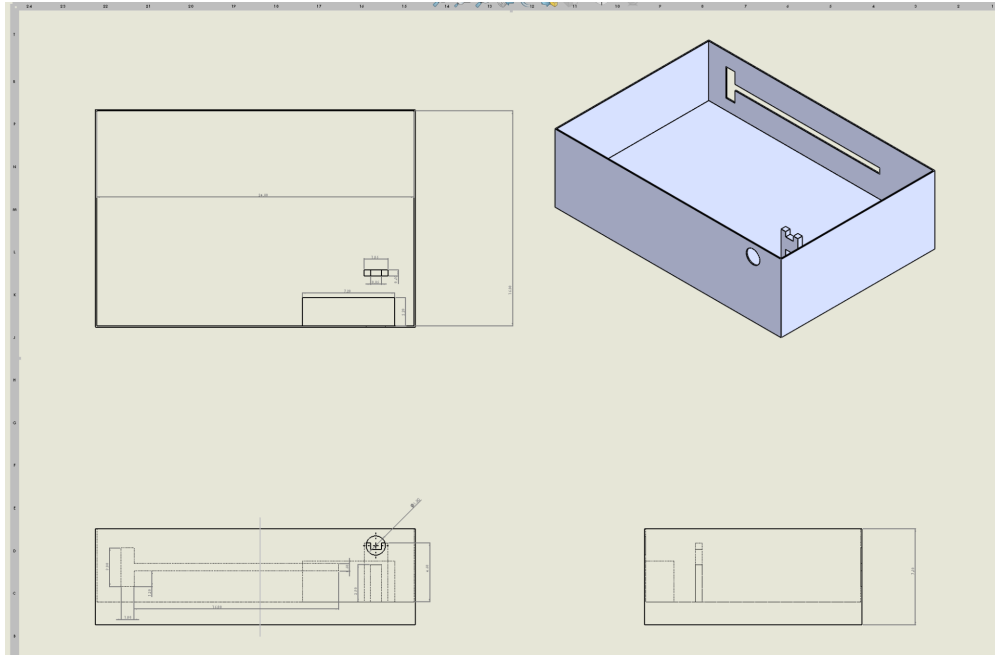
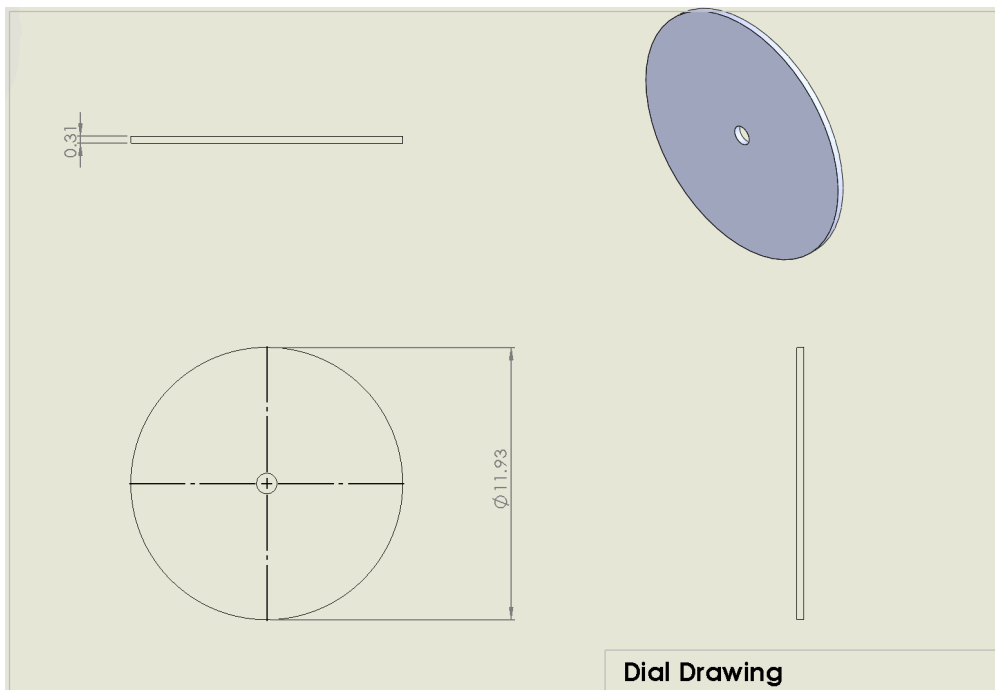


Figure 13 SolidWorks Drawing of base (3:2)



Dial Drawing

Figure 14 SolidWorks drawing of Rotary Dial (3:5)

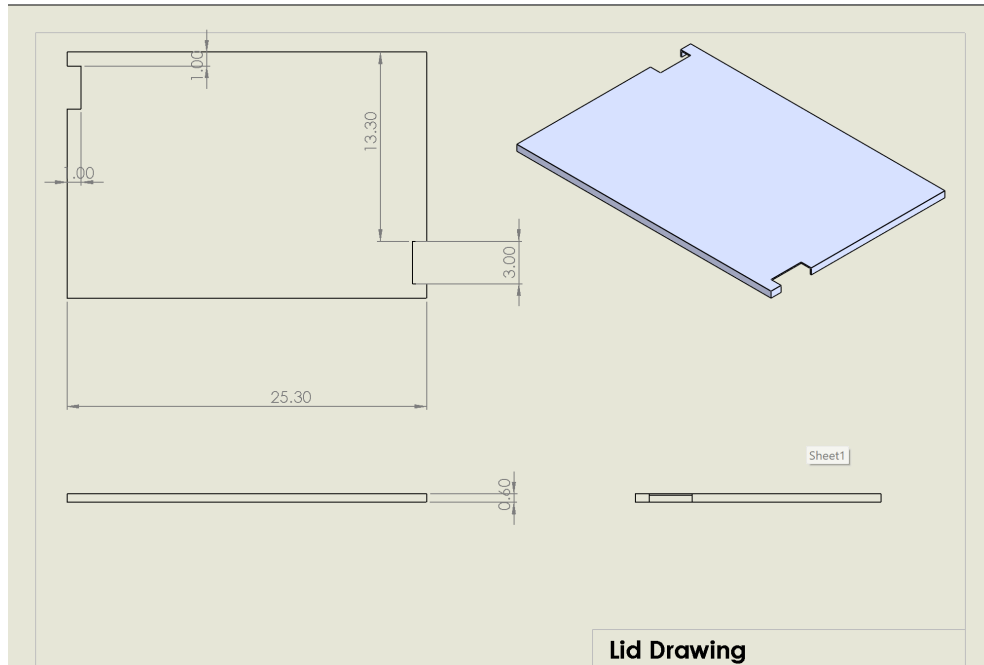


Figure 15 SolidWorks Drawing of Lid (2:5)

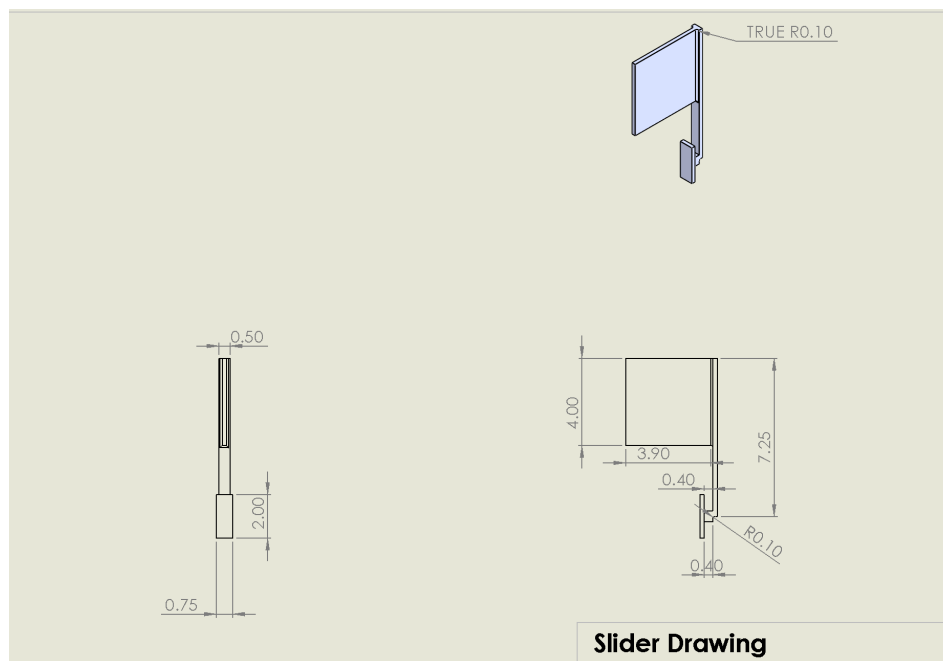


Figure 16 SolidWorks Drawing of Slider (3:5)

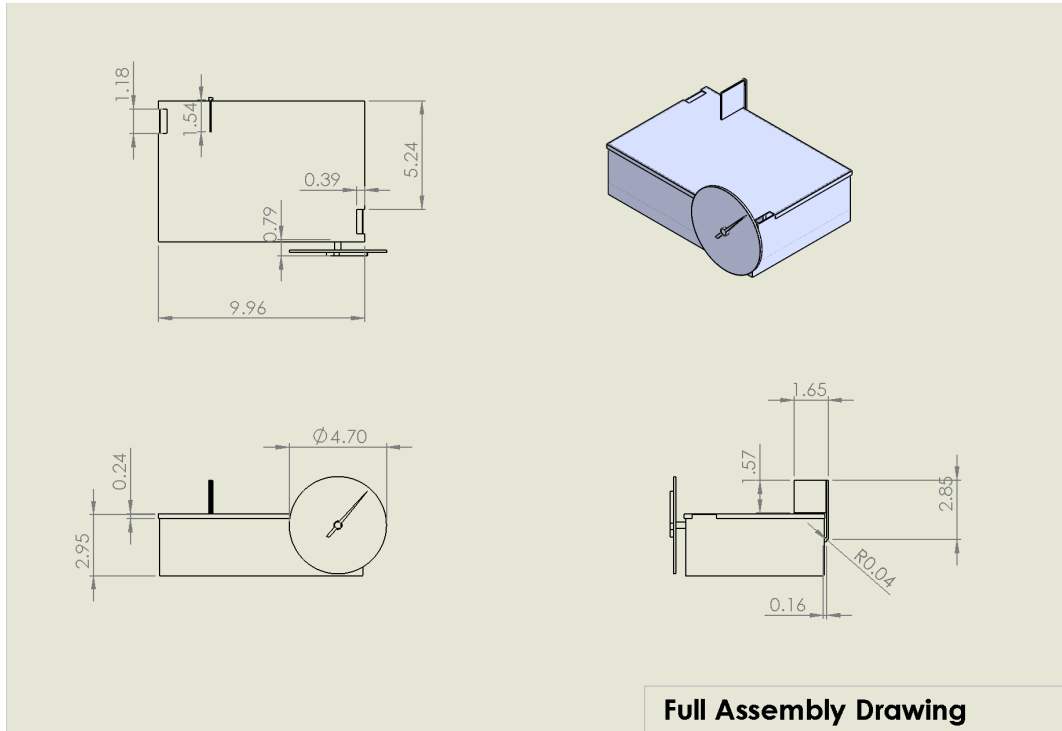


Figure 19 SolidWorks Drawing of Full Assembly (1:5)